



Qualitative test for protein and amino acid

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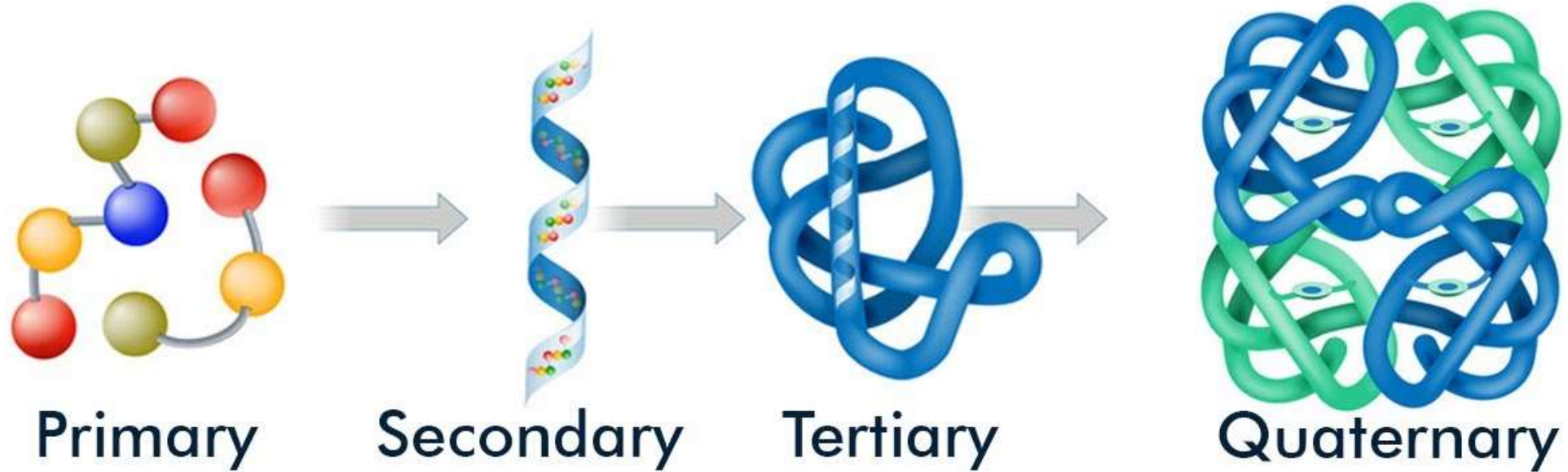
Introduction

Proteins are complex of nitrogen containing organic compounds. They constitute a major part of the living cell protoplasm, they are responsible for different functions in the cell as transport (Hb), receptor on the cell membrane, structural in bone, movement protein like actin & myosin, defensive (immuno-globulin), catalytic (enzymes) and some hormones.

Characteristics of protein:-

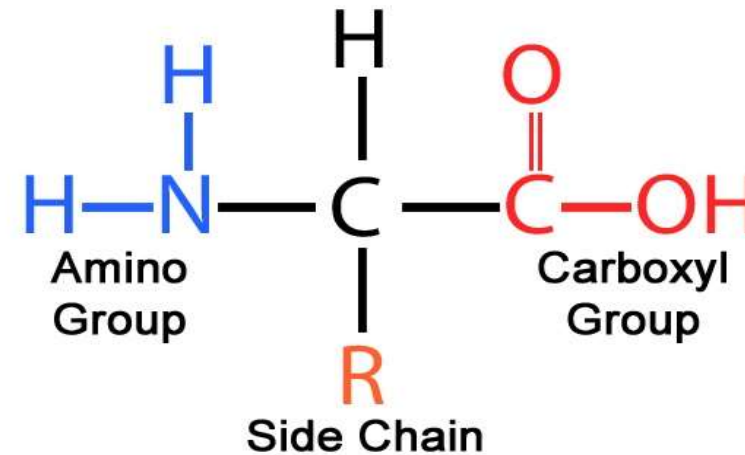
1. They are polymers of small molecules called amino acids.
2. Proteins are usually amorphous, they form colloidal solution.
3. Hydrolysis of the protein molecule by boiling with acids, strong bases and enzymes the final product being amino acids. Amino acids are crystalline and diffusible substances its solution are not colloidal and can not form any ppt.
4. Proteins exhibit amphoteric properties combining with both acids and bases to form ionizable salt.
5. Proteins can give a number of colored reactions.

PROTEIN STRUCTURE



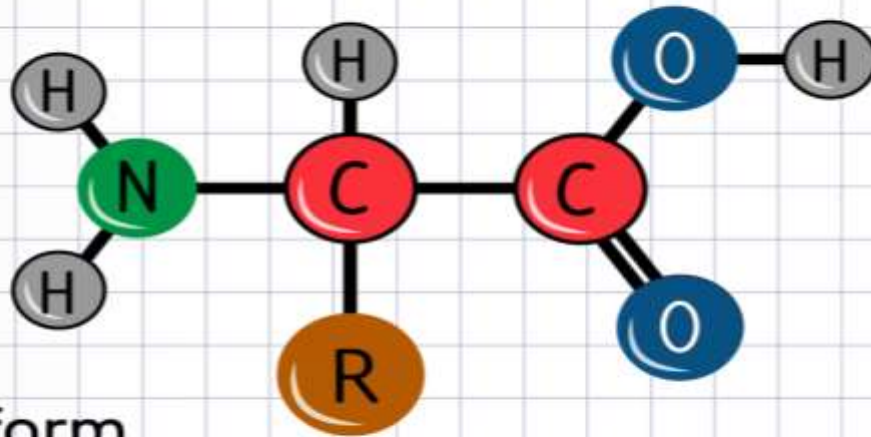
Amino acids

Amino acids are compounds that containing an amino group and carboxylic group.

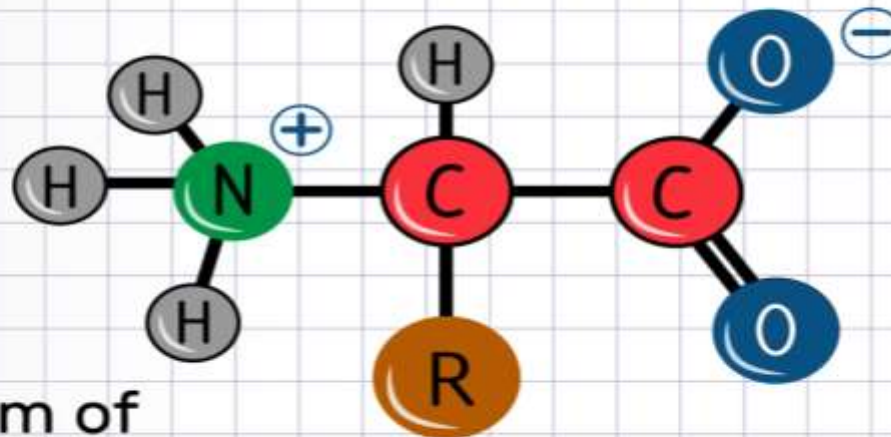
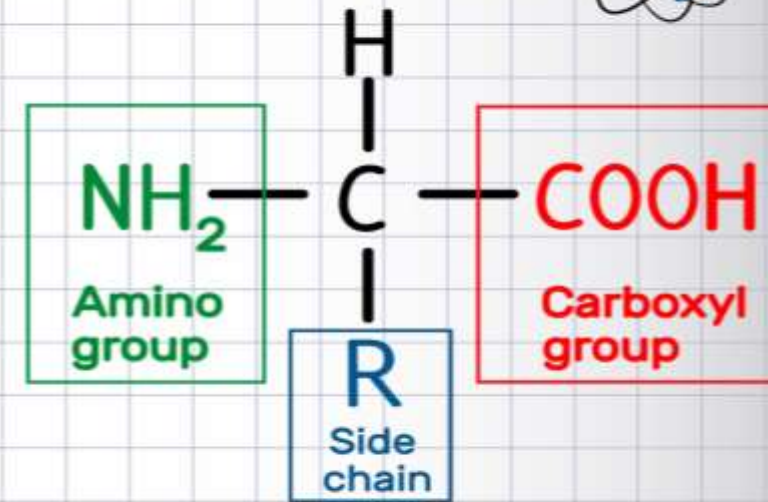


There are about 20 amino acids forms the monomer units of the protein construction. Amino acids differ from each other by there –R group.

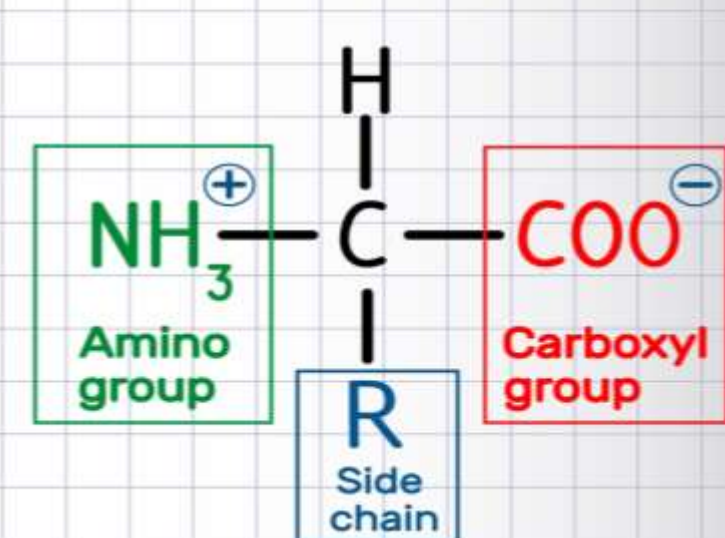
Amino acids



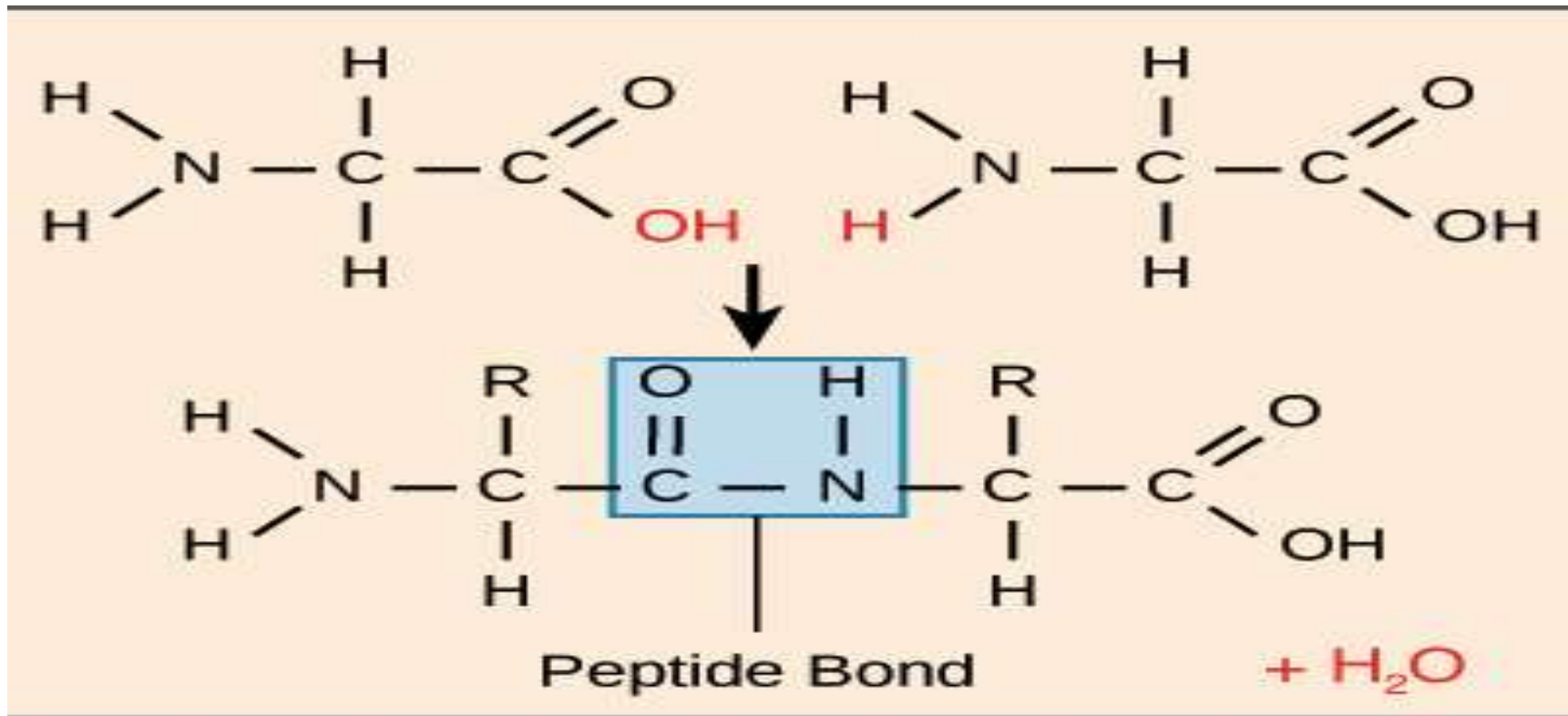
Non-ionized form
of amino acids



Ionized form of
amino acids (zwitterion)



Peptide bond :- Proteins are formed by joining the carboxyl group of an amino acid to the α -amino group of another amino acid. This bond is called Peptide bond.



Qualitative tests for proteins and amino acids

Exp. No. 1 Ninhydrin test

This test is positive for proteins and all products of protein hydrolysis including amino acids.

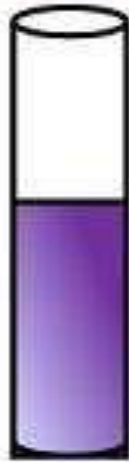
Principle:-

Ninhydrin is oxidizing reagent, oxidize the amino acids to aldehyde by forming CO_2 and ninhydrine which forms complex with other molecule of ninhydrin indicated by blue-violet color.



Procedure:-

1ml of protein solution in a test tube and add 5 drops of ninhydrin reagent, boil for 2 minutes a blue-violet color indicated protein or amino acid.

NINHYDRIN TEST	
RESULT	DETECTION
 POSITIVE: Gives purple colour	Used for the detection of all α -L-amino acids.

Exp. No. 2 Biuret test

(Distinguish between protein and amino acid)

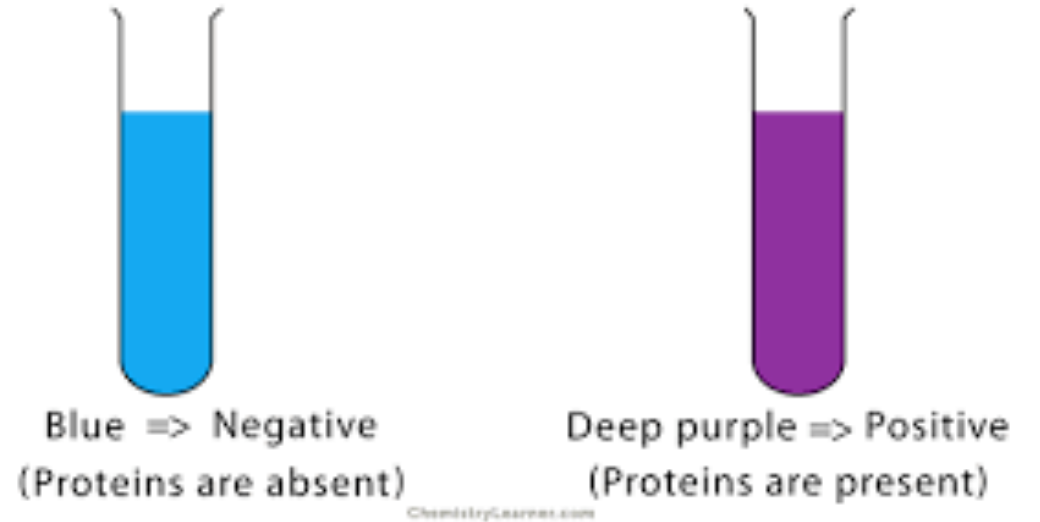
Principle:-

This test is positive with molecules contain two peptide bonds. The end reaction of the biuret test is dependent upon the formation of copper complex biuret compound.

Procedure:-

2ml of protein solution and then add a few drops of biuret reagent, a violet or rose red color appear, with peptide the color tends to pink and with more complex protein is violet.

Biuret Test Result



Exp. No. 3 Xanthoprotic test

(test for aromatic amino acids)

Principle:-

This reaction is positive in the presence of the phenyl group (C_6H_5) in the protein molecule, the positive test indicates the presence of tyrosine, phenyl alanine and tryptophan.

Procedure:-

2ml of protein solution or aromatic amino acids add 1ml of concentrated HNO_3 a white precipitate will be formed, upon heating turns yellow and finally the precipitate dissolves in the solution to produce yellow color.



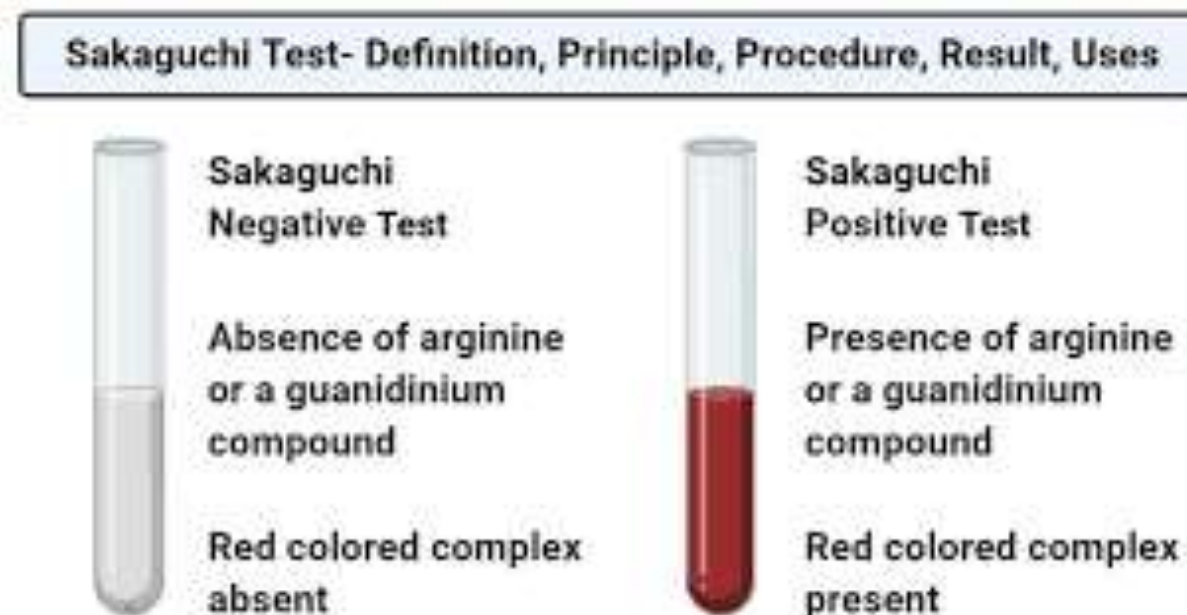
Exp. No. 4 Sakaguchi test

Principle:-

This test is very sensitive for free and combined arginine. Since all known proteins contain sufficient amount of arginine to give the positive reaction.

Procedure:-

2ml of protein solution add 8drops of 10% NaOH then add 2drops of 1% ethanolic α -Naphthol then add (5) drops of bromine water and mix the solution after each addition a bright red color appear indicating the presence of arginine.



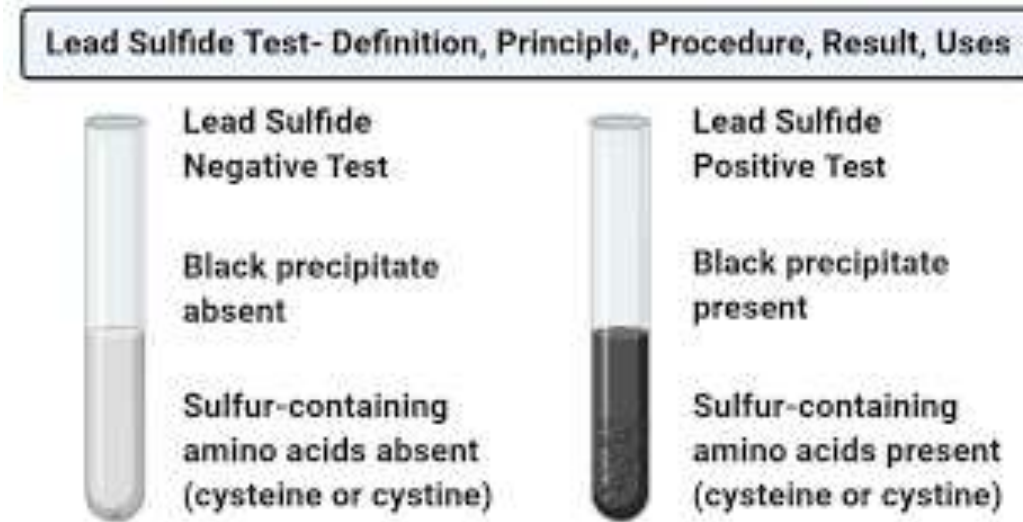
Exp. No. 5 Lead sulphide test

Principle:-

Both amino acids cysteine and cystine side chains contains sulfur group, its reaction with lead solution in basic medium produce black or brown color of PbS.

Procedure:-

1ml of protein solution add 1ml of 40% NaOH add 5drops of bromine water and produce Na_2S , then add 3drops of lead acetate solution, boil for 5 minutes a black or brown color appear due to the formation of PbS .



Exp. No. 6 Millons test

(test for hydroxy phenyl group (tyrosine))

Principle:-

Millons test is positive in the presence of hydroxyl phenyl group (HO-C₆H₅) in the protein molecule such as tyrosine.

Procedure:-

2ml of test solution add a few drops of millons reagent and mix the solution. Heat gradually to the boiling point, the precipitate or the solution or both turns to red.

MILLON'S TEST	
RESULT	DETECTION
 POSITIVE: Gives red colour	Used for the detection of Tyrosine amino acid.